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FULL RESEARCH ARTICLE

Analysis of process parameters, deposition strategies and build orientations in wire-arc additive manufacturing (WAAM)

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Abstract

Additive manufacturing (AM) has transitioned from prototyping to commercialization in many industrial sectors. In addition, direct energy deposition (DED) processes are enabling the AM for large-scale depositions. One such DED process is wire-arc additive manufacturing (WAAM), popular for its high deposition rates. The influence of process parameters and deposition strategies play a vital role in the WAAM process. Therefore, in the first part of this study, a response surface methodology (RSM) was employed based on a central composite design (CCD) for bead-on-plate analysis to understand the effects of process parameters on bead morphologies and material hardness. Similarly, for the second part of the study, different deposition strategies were employed for test wall depositions to analyse their effects on built structures. Particularly, the effects of layer-stacking on 10-layer and 20-layer walls were analysed in detail. Finally, employing an optimised deposition strategy, cubic samples were deposited with various deposition parameters based on a 2^K factorial design. Hardness, microstructural and SEM analysis were performed for in-depth analysis of the samples. Results revealed a strong relationship between process parameters (current and travel speed) versus bead characteristics and material hardness. The deposition strategies significantly influenced the built structure and wall formation. Similarly, layer-stacking played a critical role for mechanical properties of deposited samples. Cubic structures demonstrated small anisotropy for microstructural results based on various parameter settings. Similarly, small discrepancies were observed in the hardness results.

Keywords WAAM · DED-arc · Deposition strategies · Process parameters

1 Introduction

Metal additive manufacturing (AM) processes are rapidly developing owing to the particular interest from research and industry. This is particularly due to the growing need for mass customisation and resource efficiency when processing the expensive materials. Additive manufacturing processes offer such advantages compared to traditional manufacturing. Among various AM processes, powder bed fusion (PBF) [1] and directed energy deposition (DED) [2] are two important metal additive manufacturing processes. An additive manufacturing report cited by [3] shows that the total market share of PBF was significantly higher compared to DED processes, with 8.3% of market share of DED as of 2019. Surprisingly, in 2020, the share of DED among additive manufacturing processes stood at 16% [4]. This shows a significant increase in the market share of DED processes among additive manufacturing technologies. Among DED processes, wire-arc additive manufacturing (WAAM), most recently referred to as DED-arc, is gaining huge popularity among AM researchers and industries due to its many advantages. One of the advantages is the high material deposition rate. The material deposition rate of as high as 9.5 kg/h was achieved for depositing steel in a tandem WAAM process [5]. In addition, over the years, WAAM has emerged as a cost-effective AM process for medium to large parts with benefits including low material waste and shorter lead time compared to traditional manufacturing processes [6, 7].

The effectiveness of WAAM process is directly related to the process parameters and deposition strategies for various part geometries. The process parameters that significantly affect the process outcome including the wire feed speed (WFS), arc current, arc voltage, and deposition speed (or travel speed (TS)). For instance, it was shown that increasing

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the wire feed speed increases the bead width and height, while the higher travel speed decreases them [7]. Thus, the ratio of WFS/TS is the most significant process parameter for controlling the heat input where heat input influences the bead dimensions including width, height, penetration depth, penetration area, and reinforcement area [8]. Similarly, in another study [9], it was shown that apart from the bead characteristics, the process parameters also influenced the hardness of ER316Si steel. The corresponding changes in hardness were associated with the microstructural changes induced as a result of varying heat input when using different combinations of process parameters. Other studies also showed that wire feed speed, travel speed, arc current and voltage cumulatively influence the heat input which in turn effects the material properties [10–12]. Therefore, it is crucial to understand the influence of these parameters from bead-on-plate analysis. Such studies should be backed by statistical approaches requiring smaller number of experiments. Some researchers have employed useful approaches employing ANOVA or Taguchi methods to study the influence of process parameters [13–16]. However, Response Surface Methodology (RSM) with a Central Composite Design (CCD) could be more accurate and effective compared to Box-Behnken and Taguchi models involving nonlinear interactions and quadratic effects, exploring a broader experimental range with axial points, and provides better optimization and predictive capabilities, even with smaller datasets. Therefore, the first part of our study discusses the bead-on-plate trials with a statistical model based on a RSM approach.

Moreover, another important aspect of WAAM process is the deposition strategies for multi-layer deposits. The deposition strategies define the toolpath with which the welding torch traverses to deposit each layer. In case of single-pass multilayer parts, the toolpath planning is crucial, where it decides the start and end positions of the welding torch for each layer. It also includes the positions at which the arc remains energized or waits to achieve a specific inter-layer temperature or to reach to a specific position to start a new layer. For single-pass multilayer walls, it is possible to either start or end the deposition of the layers at the same positions (unidirectional deposition) or to simply reverse the direction for each layer (bidirectional deposition). It was shown in the literature that unidirectional deposition strategy causes material accumulation on the starting position while leading to material reduction at the end of the wall [17]. They demonstrated that optimal combinations of wire feed speed and travel speed could also be used to control the underlying issue. However, this can easily be resolved by employing a bidirectional deposition strategy [18]. Deposition strategies not only influence the build orientation but also the material properties. J. Wang et al. investigated different deposition strategies for titanium alloy Ti-6Al-4 V including a bidirectional strategy without any interlayer time, a unidirectional strategy with an interlayer time of 24 and 120 s, respectively [19]. Such strategies influenced the macro/microstructure of the deposited walls as well as the mechanical properties. As seen from the above literature and some other studies [20–22] that have discussed the influence of deposition paths including the unidirectional and bidirectional deposition strategies; however, a comprehensive analysis still lacks in the literature, covering a combined and comparative analysis of most influential deposition strategies. The more specific aspect in this regard, which has not been discussed widely in the literature, is the burn-back time (also referred to as dwell time). This approach resolves the issues related to the unidirectional deposition strategy by depositing more material at the end of each layer. Thus, in this study, we have studied this aspect on the wall formation and compared the results with other deposition strategies to fill the knowledge gap.

In addition, a very important aspect of large-scale deposits is the layer-stacking. This refers to the number of layers deposited layer-by-layer to achieve a large structure. During WAAM process, the subsequent layers result in partial remelting of previously deposited layers, resulting in a tempering effect on previously deposited layers. Therefore, understanding this phenomenon is essential to see how the walls deposited with different number of layers have been influenced by cyclic heating and cooling of WAAM process. Unfortunately, this phenomenon is unclear in the literature and needs a detailed study. Results shows the influence of different heat inputs [23] and interlayer temperatures [24, 25]. Nonetheless, the effects of layer-stacking on the walls remains unexplored. Therefore, in this study, the walls deposited with 10 and 20 layers were also analysed in detail to fill the knowledge gap.

Furthermore, WAAM process is desirable for large-scale depositions that involve multiple beads and multiple layers to obtain the final geometry. The high heat input involved in the WAAM process results in heat accumulation when depositing large structures. Therefore, for efficient thermal management, it is crucial to adopt the optimised deposition strategies and process parameters to avoid excessive thermal defects and geometrical inaccuracies and the corresponding mechanical properties. In multi-bead and multilayer structures, an overlap distance (hatch distance) is used between the adjacent beads. Several bead overlapping models have been studied and proposed in the literature [26–29]. An appropriate hatch distance combined with optimised process parameters can result in a smooth surface and desirable properties. On contrary, inappropriate combinations of hatch distance and process parameters can lead to valleys between the adjacent beads and may also affect the material properties. Similarly, the path planning also plays a significant role and requires an optimal deposition strategy for homogeneous

heat distribution and structural formation, as discussed in the previous paragraph. One study showed that layering sequence in additively manufactured parts effects the anisotropic behaviour of material, effecting the grain growth size and direction [30]. They also concluded that the layering only in one direction results in columnar grains for Inconel 825 alloy which was refined by a zigzag layering strategy. Thus, the layering sequence for large structures significantly influences the formability and material internal properties. Different deposition strategies have been studied in a study [31] utilizing a maraging steel for bulk deposit in cold metal transfer (CMT)-WAAM process. It was found that an oscillation strategy was not suitable for maraging steel in which local heat accumulation resulted in small shrinkage holes on the deposit. The parallel strategy was found to eliminate such holes; however, the bead shape became tall and narrow leading to lack-of-fusion defects. Nevertheless, they concluded that weaving of a single bead is an effective strategy for bead shape improvement and a reduced contact angle, leading to a defect-free overlapping with better surface finish. Moreover, apart from external features of the bulk structures, processing condition and deposition strategies can also influence the microstructure [32]. In a FEM study with five different deposition strategies for a low-carbon martensitic steel, it was concluded that phase transformations significantly affect the residual stresses and distortions in continuous patterns such as zigzag, in–out and out-in [33]. However, in line-type discontinuous patterns such as alternate and raster, phase transformations have the least influence. Alternate line pattern resulted in more uniform residual stress and distortion, making it the preferred choice. The above studies from literature show interesting strategies and improvements for bulky deposits. Nevertheless, such studies do not discuss the process outcome on several bulky deposits by employing different parameter settings. This requires several experimental trials and resources. But to realise such influences, our study used an optimised deposition strategy for cubic structures and deposited 8 cubic structures with different combinations of process parameters to witness their effects on process outcome.

Summarising the above, WAAM still lacks a systematic and comprehensive research covering the entire process chain; from parametric study based on a RSM approach to thin walls and bulky structures, analysing the influence of various deposition strategies and orientations on wall formation and material properties. Therefore, this study aims to provide a comprehensive investigation, starting with a parametric study based on bead-on-plate trials using a statistical approach. It then extends to walls deposited using single-pass multilayer and multi-pass multilayer structures, respectively. For single-pass multilayer test walls, three deposition strategies are evaluated to assess their influence on wall formation. The effect of layer-stacking is analysed for 10- and 20-layer walls. Finally, multi-pass multilayer samples are deposited with an optimized strategy, varying the arc current, travel speed, and hatch distance and the results are compared.

2 Experimental procedures

This study is primarily comprised of three parts: in the first part, a bead-on-plate analysis was conducted. For the second and third parts, thin walls and cubic samples were analysed based on various deposition strategies and build orientations. The term deposition strategy refers to the specific paths, sequences and approaches with which the torch moves to deposit layers while with build orientation in this study we refer to the spatial arrangements such as walls deposited using different number of layers or the difference between thin walls and cubic structures. Moreover, the thin wall in WAAM deposits typically refers to walls deposited employing a single pass.

In the first part of the study, a bead-on-plate analysis was conducted to analyse the effects of arc current and travel speed on bead width, bead height, penetration depth and hardness. The bead-on-plate study was based on a statistical approach using a response surface methodology (RSM) by employing a central composite design (CCD) in Minitab software. The RSM model requires the maximum and minimum values of factors under study which then creates a structured experimental design, fitting a polynomial regression model to explore relationships or optimize the responses within the studied range. The maximum and minimum values of arc current were 80A and 200A while for that of travel speed 7 and 20 mm/s were used. The selection of the parameter range was based on prior trials. The deposition current below 80 A resulted in lack of fusion while above 200 A increased the heat input significantly. Thus, to maintain a stable deposition within moderate heat inputs, the underlying parameters were selected for this study. The selected travel speeds ensured a low, medium and high level of responses variables, helping in deeper understanding of the combined effects of these parameters. The full experimental design for bead-on-plate analysis is shown in Table 1. Finally, using the experimental design, single pass beads were deposited from each parameter combinations which are shown in Fig. 1. ER70S-6 steel wire with a diameter of 1.2 mm was used for depositing all the samples using a GMA-based WAAM system operated with a synergic control in direct current electrode positive (DCEP) polarity. The heat source (Selco Genesis 352 GSM) was integrated with a wire feeder (Selco WF104) to precisely feed the wire into the melt-pool. A 6-axis Yaskawa Motoman HP20 robot was used for path-planning. A mild steel plate was used as substrate. To protect the melt-pool during deposition, pure

Table 1 Process parameters for bead-on-plate analysis

Sample	Std Order	Run Order	Current [A]	Travel speed [mm/s]
S1	8	1	140	20
S2	5	2	80	13.5
S3	1	3	97.6	8.9
S4	7	4	140	7
S5	10	5	140	13.5
S6	3	6	97.6	18.1
S7	9	7	140	13.5
S8	13	8	140	13.5
S9	12	9	140	13.5
S10	11	10	140	13.5
S11	2	11	182.4	8.9
S12	6	12	200	13.5
S13	4	13	182.4	18.1

argon + 20% CO_2 was used at a flow rate of 18 l/min. After the deposition, all the samples were cut and resin mounted and polished using abrasive papers with grit size ranging from 80 to 2500. In addition, the samples were polished with diamond spray of 6 and 1 microns, respectively. Macro and microstructural analysis were performed using a macroscope (Vert.A1 Ziess Axio) and optical microscope (Carl Ziess Stemi 508 doc), respectively. Hardness tests were performed on a Vickers hardness tester for a load of 1 kg and an indentation time of 15 s. The data of bead morphologies were obtained by accurately measuring the bead width, height, and penetration depth from macroscope. All the data were transferred to Minitab where further analysis was performed that are discussed in Sect. 3.1.

For the second part of this study, single-pass multilayer walls were deposited with 150A current in pulse transfer mode and 11 mm/s travel speed. From our prior experience, these parameters proved to generate a more stable arc, better wall geometry and generate moderate heat input, avoiding significant wait time between layers. Each wall was deposited with a length of 50 mm and consisted of 10 layers. Three deposition strategies: unidirectional, bidirectional and unidirectional with dwell time were used to deposit 3 walls to analyse the effects of these strategies on wall formability. In the unidirectional strategy, each subsequent layer was deposited in the same direction. After depositing a layer, the welding torch (with the arc off) was moved back to the starting position. The process was repeated with an increment in the z-direction equal to the layer height for each new layer, continuing until the 10th layer was deposited.

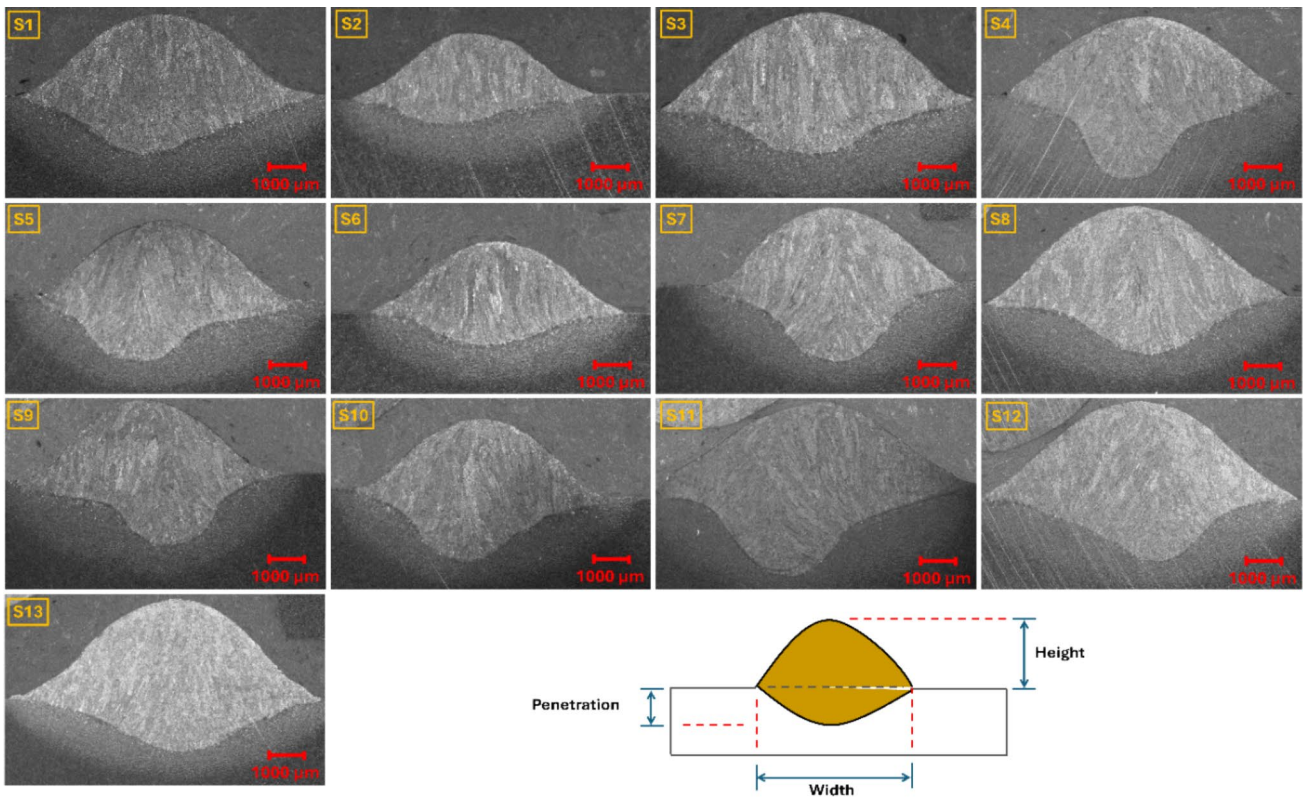


Fig. 1 Bead sections for samples deposited under different combinations of arc current and travel speed

In the bidirectional strategy, once the layer had reached the end of the wall length, the torch was moved upward in the z-direction by an amount equal to the layer height and then reversed its direction to deposit the subsequent layer. Finally, for the third strategy, the same unidirectional path was used but at the end of each layer a dwell time (also referred as burn-back time) was added which kept the arc energized for a small time instant to melt and deposit more material at the end of each layer while the robot had stopped.

In addition, WAAM process uses a layer wise approach to deposit large structures. The layering is followed by alternative thermal cycles. The continuous deposition results in heat accumulation and corresponding tempering effects. This becomes pronounced as the number of layers increase on the deposited walls. Therefore, a detailed analysis is necessary to understand the effects of layer-stacking. Thus, for this study, two walls were deposited with same process parameters in bidirectional deposition strategy. However, the two walls were composed of different number of layers to assess the influence of layer-stacking on the properties of deposited walls. The walls comprising of 10 and 20 layers were analysed for hardness, microstructure and under scanning electron microscopy (SEM).

In the last part of the study, eight different multi-pass multilayer cubic structures were deposited with an optimised deposition strategy in which an outer shell is deposited with a single pass as the perimeter of the cube, followed by infill passes deposited with different hatch distances. A 2^k factorial design was used with 3 factors including the arc current, travel speed and hatch distance, thus, 8 cubic samples were deposited. Table 2 reports the process parameters for these samples and this range of parameters was selected to avoid wall collapse and ensuring stable deposits while providing a good combination of parameters to provide significant changes in the response variables.

Table 2 Process parameters for cubic deposits

Sample	Current [A]	Travel speed [mm/s]	Hatch distance [mm]
C1	120	8	2
C2	120	8	3
C3	120	15	2
C4	120	15	3
C5	160	8	2
C6	160	8	3
C7	160	15	2
C8	160	15	3

3 Results and discussion

3.1 Parametric study

The statistical analysis revealed interesting relationships between the process parameters and the response variables. The ANOVA table from RSM regression is shown in Table 3 with regression equations for each response variable. The arc current and travel speed have very low *P*-values with a 95% confidence interval, suggesting their statistical significance for width, height, penetration and hardness. In addition, the R^2 and adjusted R^2 show high values, confirming the robustness of RSM model. Similarly, the *P*-values above 5% for lack-of-fit indicate that the model fits the data.

The main effects of the process parameters and their interaction effects on bead width, height, penetration depth and material hardness are illustrated in Fig. 2. Looking at the main effects plots, we can observe a significant effect of arc current on bead width, height and penetration depth. With increased current, all the bead characteristics tend to increase, showing a positive relationship. An increase in arc current increases the wire feed speed and more material is fed into the melt-pool.

Thus, the increased melting of the feedstock material with higher arc current creates a bigger melt-pool, leading to higher bead width and reinforcement height. Similarly, the higher current causes a deeper remelting of the substrate, resulting in deeper penetration. On contrary, travel speed

Table 3 Analysis of variance for factors and response variables with regression equation for each

Factors	Responses	R^2	R^2_adj	*P*-value	Lack of fit *P*-value
[Current], [Travel speed]	Height	89.49%	87.38%	[0.000], [0.007]	0.754
[Current], [Travel speed]	Width	90.82%	88.98%	[0.000], [0.000]	0.554
[Current], [Travel speed]	P. Depth	80.29%	76.35%	[0.001], [0.002]	0.07
[Current], [Travel speed]	Hardness	86.03%	83.23%	[0.000], [0.000]	0.101

Height = 0.713 + 0.01273 Current – 0.0468 Speed

Width = 3.933 + 0.03597 Current – 0.1965 Speed

P. Depth = 0.989 + 0.01264 Current – 0.1001 Speed

Hardness = 297.3 – 0.4709 Current + 4.384 Speed

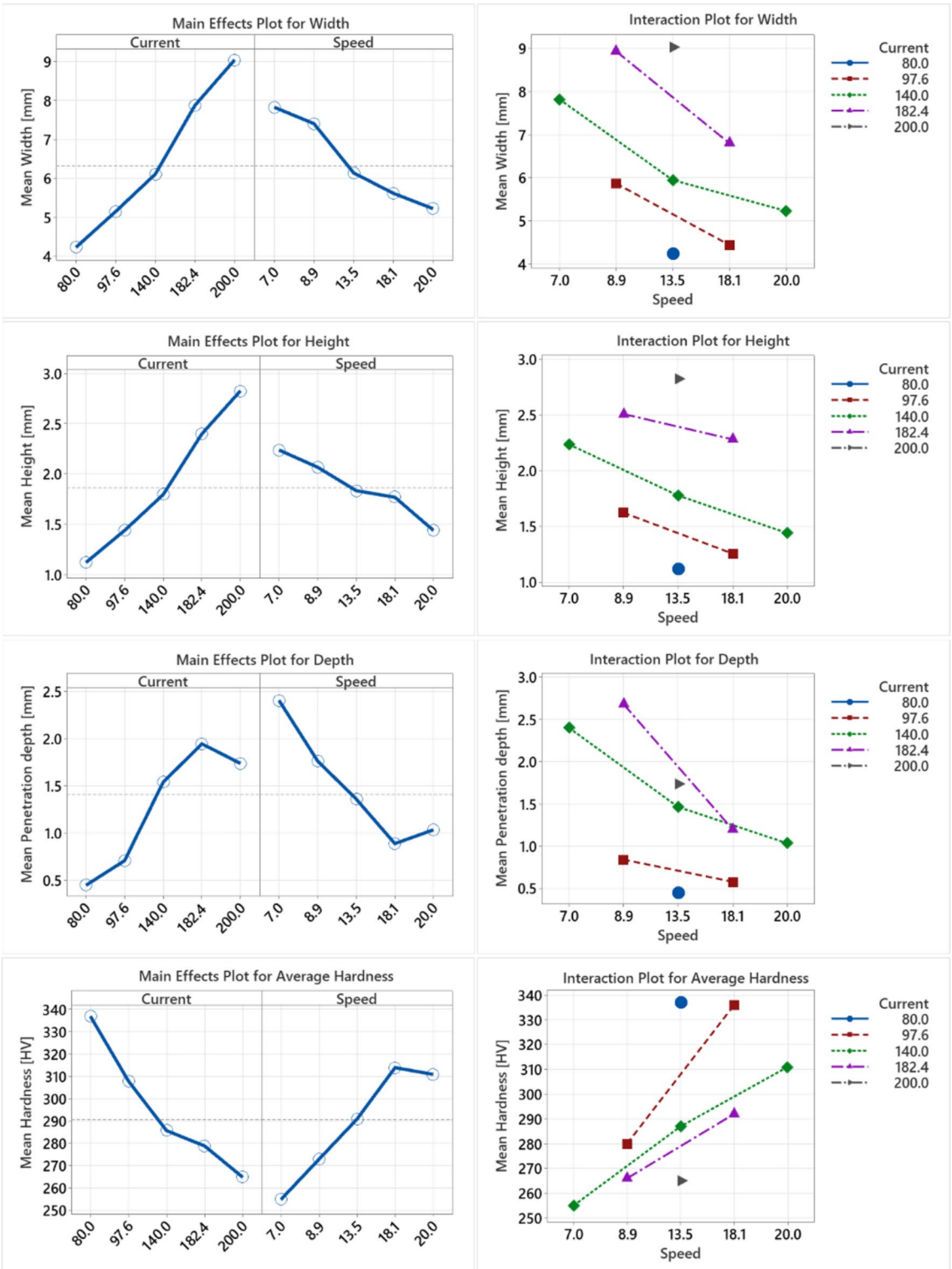


Fig. 2 Main effects and interactions of current and travel speed on bead width, height, penetration and material hardness

exhibits an explicitly inverse relationship with bead width, height, and penetration depth. This is associated with the reduced time available for material deposition; decreasing the amount of molten metal per unit area as the welding torch moves faster. Literature shows similar results for 316L even employing a different statistical approach [34].

Unlike the bead morphologies, material hardness responds differently to the changes in current and travel speed. As the current increase, the hardness seems to decrease significantly, indicating that the higher current resulting in higher heat input possibly softens the material under deposition. Conversely, higher travel speeds lead to higher material hardness. One possible explanation for this behaviour can be associated with the fact that higher heat input may lead to larger grain size, resulting in lower hardness. Similarly, as the travel speed increases, the width of the melt-pool decreases, leading to faster cooling. The faster cooling usually causes finer and smaller grain size which results in higher hardness.

In addition, the interaction plots further explain the combined effects of current and travel speed on the response variables. The plot lines which intersect with each other on the interaction plot represent a significant interaction of process parameters for that response variable. The interaction plots reveal that the effect of arc current on bead characteristics is more evident at lower travel speed. However, at higher travel speeds, the influence of current on these outcomes become less pronounced, where lines tend to converge at various current. Hence, it would not be incorrect to say that at higher levels, travel speed seems to be dominant than arc current. Moreover, the interaction of current and travel speed for hardness shows that material softening at higher current is more evident at lower travel speed. The material hardness tends to increase with higher travel speeds at all current levels, indicating that the heat induced softening of material can be mitigated with higher travel speeds.

The contour plots illustrating the influence of arc current and travel speed are included as electronic supplementary material, shown in ESM. 1, representing their effects on bead width, height, penetration depth, hardness, dilution and ratio of heat effected zone to fusion area (HAZ/Fusion). The bead width shows interesting results in which it grows wider at higher current, particularly at lower speeds. Particularly, the bead widening of above 10 mm occurs at around 200 A current and travel speeds below 12 mm/s. The increasing travel speed decreases the bead width even at higher current levels. Similarly, a bead height of more than 3 mm can be achieved at arc current of 200 A and travel speed of 8 mm/s. The higher travel speeds significantly reduced the height, indicating that at lower speeds more material accumulates to produce taller beads while higher travel speeds reduce the amount of deposited material per unit area, leading to shorter beads. In case of penetration depth, the contour plot follows a comparable trend to width and height, where high current and low speed seem to achieve the deepest penetration, exceeding 2.5 mm. result. As the travel speed increases, a shallower penetration is achieved.

The contour plot for average hardness shows that a very high hardness above 360 HV is possible at very high travel speed (around 20 mm/s) and lowest arc current (around 80 A). Looking at the plots for width, height and penetration, it is evident that this combination of current and travel speed produced the lowest values for these variables. This indicates very small bead dimensions, which suggests a relatively fast cooling, leading to high material hardness. Moreover, the plot suggests that the lowest average hardness is achieved below 9 mm/s of travel speed between 125 and 200 A current.

Another important variable for deposited beads is the dilution, representing the amount of re-melted material within the substrate with respect to the reinforcement area above the substrate. In additive manufacturing a lower dilution is desirable because severe re-melting in the layer-by-layer process may influence the material properties of previously deposited layers. However, the penetration should be sufficient to ensure the structural integrity. A similar indicator is the HAZ/Fusion where a higher ratio indicates more heat accumulation and less material deposited above the substrate. The higher heat accumulation can influence the mechanical properties. The results indicate that proper tuning of current versus travel speed is necessary to achieve a desirable dilution since various combinations of these parameters result in varying degrees of dilution. For instance, between 125 and 200 A current and below 14 mm/s, the dilution can exceed 40%. The proportion of reinforcement area with respect to re-melt area can be increased (achieving lower dilution) by feeding more material for a specific arc current. However, most of welding sources used for DED-arc processes work on a synergic operation. Therefore, any increase in wire feed speed results in higher arc current. Nevertheless, de-coupling the wire feed speed to that of arc current (non-synergic operation) can resolve this issue by feeding more material at a specific current so that most of the heat generated by the power source can be utilised for melting the extra material than transferring in the substrate. Another way to resolve this issue in the synergic operation is to use another cold wire that is fed independently without effecting the arc current, while the current can be controlled by the wire feeding of primary hot wire, as seen in the literature [35]. The results show that a lower HAZ/Fusion ratio can be achieved by increasing both the current and travel speed, however, the various ranges of parameters influence the outcome. Hence, proper tuning of these parameters can produce desired outcomes.

A three-dimensional visualization of the relationships between arc current, travel speed, and the corresponding outcomes are illustrated in supplementary materials, ESM. 2. The width exhibits a sharp upward trend as the current approaches 200 A, especially when the travel speed approaches between 8 and 10 mm/s. In applications where wider beads are desirable, higher current coupled with low travel speeds would be advantageous. Conversely, the bead

width is highly influenced by travel speed where high travel speeds results in narrower beads even at higher current levels. Similarly, the bead height represents a comparable relationship to travel speed and current as that of the bead width. The underlying results from the surface plot suggest that one should operate at a higher current and lower speed to achieve the maximum height while the higher travel speed results in the reduction of the vertical bead buildup. The steep surface plot of penetration explicitly shows a strong correlation between the penetration depth and arc current in which the higher current and slower travel speed causes a higher energy input to penetrate deeply into the substrate. Confirming the results from previous visualisations, the surface plot of hardness exhibits a downward slope when current is increased at lower travel speeds. However, increasing the travel speed moves the plot upward, showing higher hardness. Moreover, the dilution exhibits a more complex relationship. Initially, it increases with the increase of current reaching to the peak at around 150 A current and then exhibits a more intricate correlation on the influence of process parameters. To control the dilution, proper attention must be given to the combined effects of current and travel speed at various levels. The surface plot of HAZ/Fusion ratio adds a 3D visual of previous results, demonstrates a complex relationship at various parameter levels; however, it clearly shows that a lower HAZ/Fusion ratio is achievable at high current and high travel speed.

3.2 Deposition strategies for walls

In layer-by-layer AM processes based on direct energy deposition (DED), the choice of deposition strategy plays a very important role in determining the wall formation, orientation and build dimensions. These deposition strategies influence the alignment, thickness uniformity and surface evenness of deposited walls. Therefore, this part of study analyses the effects of different strategies on physical orientation and formation of deposited walls using a unidirectional strategy, a bidirectional strategy and a bidirectional strategy with an added dwell time at the end of each layer.

Figure 3 shows the walls deposited under unidirectional, bidirectional and unidirectional with dwell time from top to bottom. The unidirectional strategy resulted in a very pronounced angular deviation in the Z-direction when each successive layer was deposited. The figure shows that this deposition approach creates a wall that is thicker at the end where the deposition is started and gradually tapers towards the end, creating a wall with an explicit slope. Thus, this deposition strategy possesses huge challenges for structures where vertical alignment and surface smoothness are critical for overall performance.

In contrary, when the deposition direction was reversed for each layer by adopting a bidirectional deposition strategy, it was found to be very effective. This alternating pattern helped in counterbalancing the forces causing a slop resulted

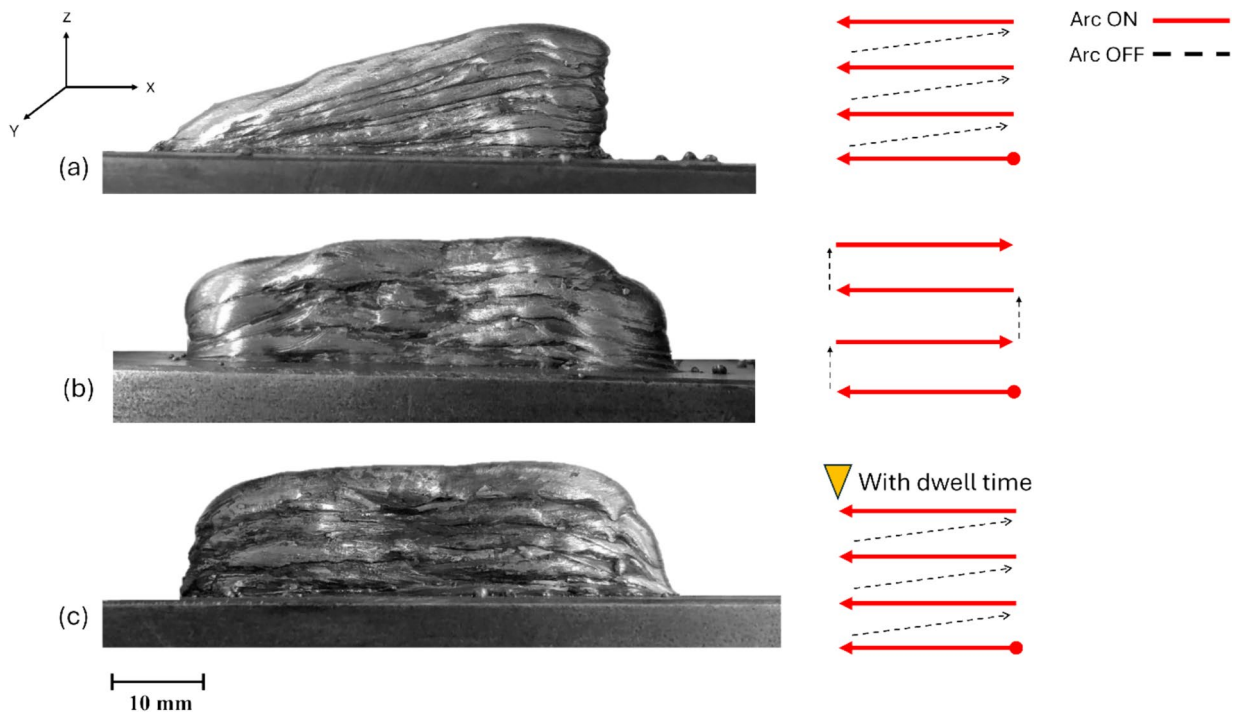


Fig. 3 Single-pass multilayered walls deposited with same parameters using, **a** unidirectional strategy, **b** bidirectional strategy, and **c** unidirectional strategy with dwell time

from unidirectional strategy, leading to a wall showing more vertical alignment at the two wall edges with uniform height and a smooth surface finish. Similar results have been reported in the literature for other DED processes using the unidirectional and bidirectional deposition paths [36, 37]. Nevertheless, the bidirectional deposition strategy exhibits a very slight angular slip at the both ends of the wall which is unavoidable in the DED-arc process. This can be machined if a very flat surface is necessary, however, on the expense of a higher cost by resulting machining and material waste.

Moreover, it was noticed that the unidirectional deposition strategy with dwell time modifies the standard unidirectional approach by keeping the arc energized for a short period at the end of each layer. The additional dwell time allows for melting and depositing more material to be deposited at the end of each layer, helping in mitigating the slopes observed in standard unidirectional deposition strategy. Adding a dwell time proved to be very effective to counter the issue of wall formability in unidirectional path strategy, thus a careful control of the dwell time improves the vertical alignment by reducing the tapering effect of the wall from one side to the other. However, the effectiveness of this strategy lies in precise control of dwell time duration. When adding a dwell time, its control is essential; if kept too short duration, the arc would turn-off before the length of the layer is reached, leading to issues of wire sticking with the melt-pool upon solidification. If kept too high, it can lead to burning the nozzle tip. Thus, a proper dwell time duration is essential.

In short, the choice of deposition strategy significantly influences the wall orientation, formability, and surface smoothness in the DED-arc process. The unidirectional strategy is simple to implement but presents challenges in achieving the desired wall geometry. The bidirectional strategy is practically more useful for obtaining smoother surfaces without significant concerns. The addition of dwell time in unidirectional path strategy helps in achieving the desired geometry, however, careful attentions should be kept for avoiding unwanted outcomes.

3.3 Influence of layer-stacking in thin walls

To realise the influence of layer-stacking, the walls consisting of 10 and 20 layers were deposited, respectively. Figure 4 shows the macroscopic cross-sections, providing a visual of the two walls. These walls reveal some noticeable differences that can be attributed to the number of layers used during deposition.

The wall deposited with 10 layers exhibits a more consistent structure, relatively maintaining a uniform wall width and orientation. On the other hand, the wall consisted of 20 layers exhibits slight irregularities in layer alignment and some variations in the width. These

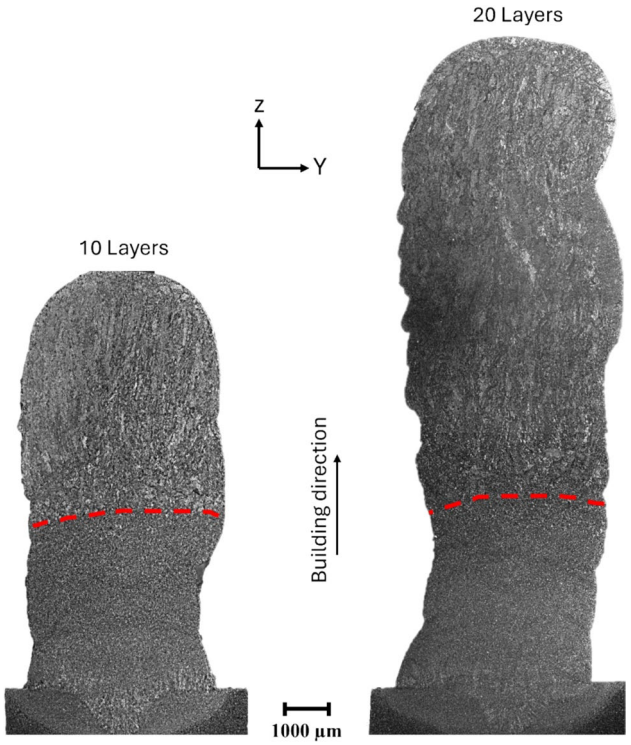


Fig. 4 Macrostructural sections for walls deposited with 10 and 20 layers

differences suggest that the increased number of layers in the building direction result in more heat accumulation, potentially leading to minor distortions. Depending on the number of layers for thin walls, proper interlayer wait time may be required to control the heat accumulation to avoid the wall distortions and corresponding material properties. This can be a compromise for production time but may ensure a consistent wall geometry and desired properties.

Apart from the dimensional considerations, a clear pattern emerges in the building direction of both walls which are separated with red dotted lines on the figure. It is interesting to note that this pattern distinguishes in the two wall sections after a certain wall height. Considering the total wall height and average layer height, it was found that this difference in the macrostructure appears at around and after 5th layer. The lower section exhibits a finer and more uniform structure. This indicates a possible equiaxed grain with finer grain size due to rapid cooling in the proximity of substrate. As the deposition progresses in the building direction, both walls display a transition to a more columnar type of structure. Similarly, the wall width tends to increase wider at the upper sections of both walls. We assume these discrepancies in walls widths and structures are associated to heat accumulation in the upper wall sections due to lower thermal path for heat transfer through conduction.

Further analyses were performed to determine the difference between the two wall sections based on layer-stacking. Figure 5 shows, on the left, the positions of hardness tests, microstructure and SEM analysis, and on the right, it details the hardness results for both walls. The hardness graphs illustrate the hardness profiles from top to bottom in the building direction and the corresponding average hardness at three distinctive locations on the walls i.e. upper, middle and bottom, respectively. Similarly, it also illustrates the overall wall average for both wall samples.

The hardness profiles of both walls, starting from substrate, show a comparable initial trend with similar hardness levels. Moving in the building direction, both walls exhibited a gradual increase. However, as the layer stacking progressed, the wall deposited with 20 layers showed an evident difference compared to the 10-layered wall, demonstrating a lower hardness profile in the upper section of the wall. This is likely due to the heat accumulation and prolonged exposure to high temperature, leading to a comparatively lower hardness profile for the wall with 20 layers.

The wall with 10 layers also shows variations in hardness in the building direction. However, it does not show the same degree of variations observed in the wall with 20 layers. Thus, it will not be incorrect to say that the wall deposited with 10 layers retains a higher hardness throughout its height, likely due to faster cooling rates and less heat accumulation.

The bar plots showing the average hardness for both walls further highlight the main results, helping to reach a meaningful conclusion. The wall with 10 layers shows that the hardness slightly increased in the building direction with an average hardness of 175 HV, 168 HV, and 154 HV in the top, middle and bottom sections, respectively. The wall with 20 layers, shows similar average hardness in the bottom and middle sections with average hardness of 149 HV and 148 HV, respectively. However, it increases in the top section with an average hardness of 171 HV. Moreover, when considering the total wall average, the wall built with 10 layers demonstrated a slightly higher average hardness compared to that built with 20 layers.

The results of optical microscopy from different locations of the walls are illustrated in Fig. 6. The top sections of both walls represent predominantly columnar microstructures with elongated grains. The columnar structure is typical in additive manufacturing where the solidification direction follows the heat flow. Since in thin wall sections, the heat transfer takes place mostly through conduction, passing through the length of the wall in the building direction and

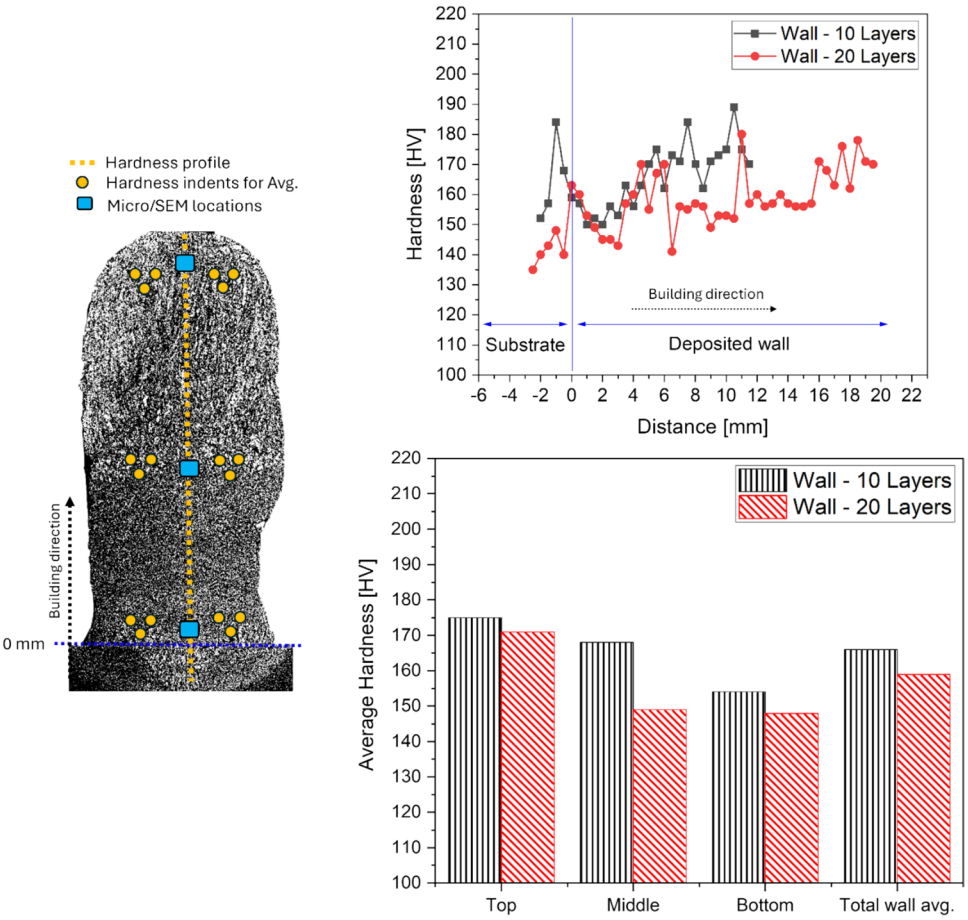


Fig. 5 On the left, shows the positions for hardness, microstructure and SEM analysis, and on the right the figure shows the hardness profiles in the building direction and average hardness at various locations and the average hardness for the entire wall

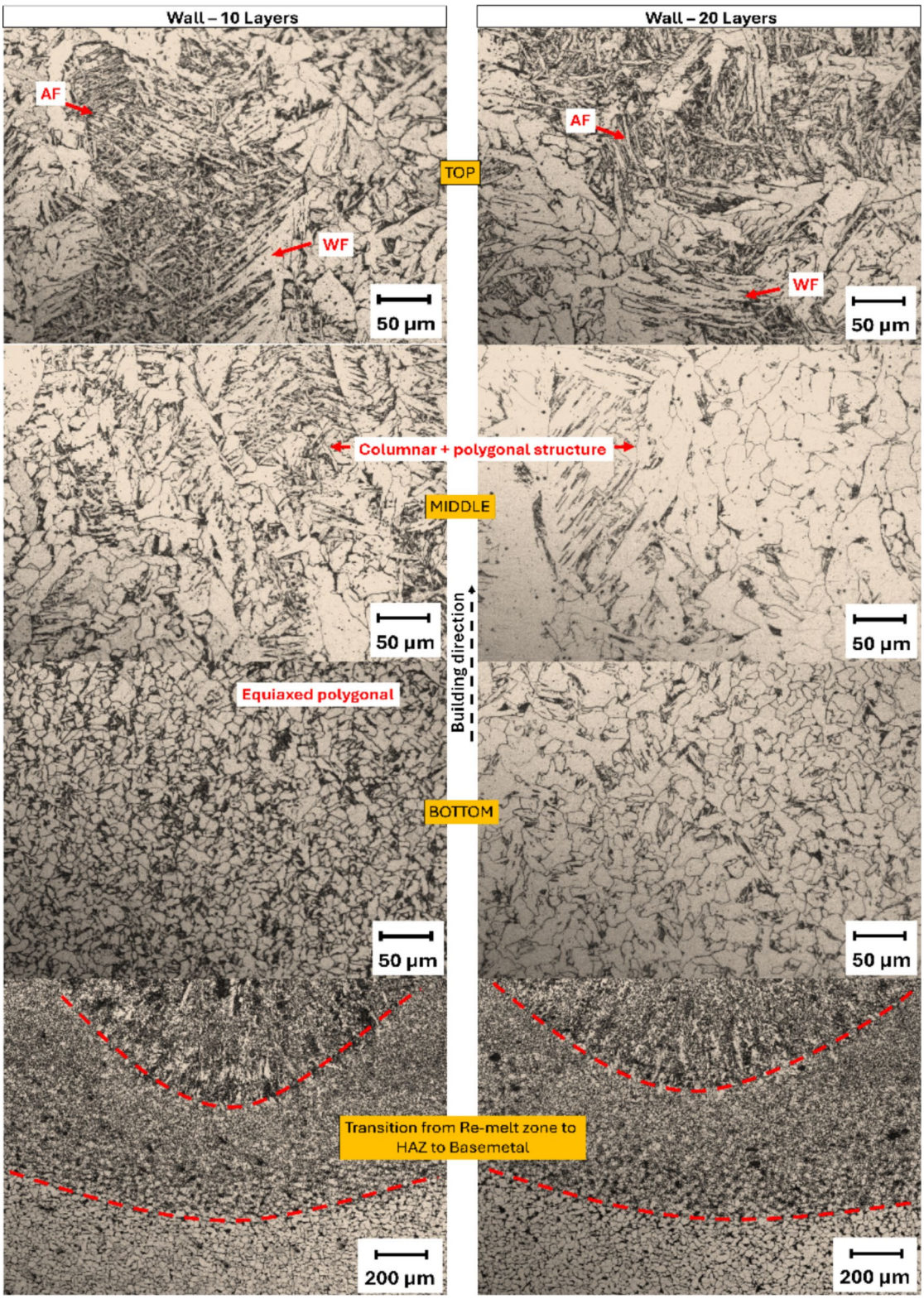


Fig. 6 The optical microscopic images of microstructure in the building direction for walls deposited with 10 and 20 layers, respectively

dissipated across the substrate. This typically results in a columnar structure, especially in the upper wall sections. In addition, the top sections of both walls show different morphologies mainly consisted of needle-shaped acicular ferrites (AF) and Widmanstätten ferrite (WF). These microstructures are typically associated with slower cooling which increases the hardness and strength without hindering the ductility [38, 39]. Thus, the upper section of the walls owing to a slower cooling favours the formation of acicular and widmanstätten ferrites.

The middle sections of both walls represent an evident transition compared to the top sections. The predominantly columnar grain structure transitioned to a mixed structure representing partly polygonal grains. The middle section of the wall deposited with 10 layers shows a finer and well-defined grain structure while the 20-layers wall shows partly elongated and coarser grains. The increase in the layer count till this point (middle section) appears to have slowed down the cooling rate sufficiently to allow the growth of larger ferrite laths.

Moving down to the bottom of the walls, a well-refined microstructure is seen. Due to the proximity with the cold substrate, acting as a heat sink, the rapid cooling of the melt-pool resulted in much finer and more equiaxed grains. However, the microstructure across the two walls in the bottom section clearly shows different grain size. The wall deposited with 20 layers represents a coarser polygonal grain structure compared to 10-layered wall. This can be associated with the layer-stacking as the higher number of deposited layers result in a more intense thermal history, higher heat accumulation and with each new layer the previously deposited material undergoes a tempering effect which influences the grain structure.

The microstructural evolution in the deposited walls characterize the thermal history of the material under deposition. The distinct grain structures and phase distribution in various sections of the walls are associated to the varied thermal gradients and cooling rates. The observed transitioning of the microstructural features from finer, equiaxed grains near the substrate to more columnar structures as we move towards the upper section in the building direction represents the dynamics of cooling rates in shaping the microstructure. Similarly, the influence of the number of layers for single-pass multilayer walls are associated with the thermal history where higher the number of layers, lower will be the cooling rate. As the number of layers or heat input increase, the average grain size increases [40]. Thus, varied thermal history results in comparable microstructural morphology, however, effecting the phase fraction and grain size across various sections of the walls depending on the number of deposited layers.

The optical microscopy provided a detailed understanding of the underlying microstructure, featuring distinctive phase structures and grain morphologies. However, scanning electron microscopy (SEM) provided additional information for the deposited walls. The SEM images of three distinctive locations of the two walls are reported in Fig. 7. The 10-layer wall represents a comparatively rough and uneven microstructure at the top. Both walls exhibit certain micro-voids or pores with varying degree of size and fraction. However, as can be seen in the SEM image of top section, the 10-layer wall shows a higher density of these voids and irregularities. This phenomenon is intriguing; can be related to the solidification processes and the corresponding thermal history experienced across the two walls based on the number of layers. The 20-layer wall shows a more homogeneous microstructure with no visible micro-defects at the top section.

Moving towards the middle section of the walls, a noticeable difference can be seen across the two wall samples. The 10-layer wall continue to show the micro level porosity, however, the there is evidence of more consistent grain structures compared to the top section. Similarly, a more homogeneous columnar grain can be observed. The wall deposited with 20 layers exhibits a more pronounced grain growth with well refined and uniform mix of columnar and polygonal grains. It also shows some voids in the middle section: less in number but larger in size compared to 10-layer wall.

The microstructures of both samples illustrate a homogeneous polygonal structure at the bottom of the walls. The 10-layer wall seems to have undergone significant grain refinement, probably due to the faster cooling with substrate less cycling heating owing to lesser number of layers. The 10-layer wall bottom section also shows some micro-defects; however, the concentration seems to be slightly less compared to the middle and top sections. Similarly, the 20-layers wall also shows a polygonal and uniform microstructure with no signs of defects.

The results demonstrate that the number of layers in thin walls significantly influence the microstructure and hardness of the deposited material. The higher number of layers reduced the risk of micropores in the deposited wall and different microstructural features were observed between the two walls. However, the hardness of the 10-layer wall found to be slightly higher than the 20-layer wall. Presumably the slightly higher hardness of 10-layer wall occurred as a result of faster cooling on the expense of producing more pore-like defects in the deposit. Thus, further experimental analyses with destructive testing are recommended to unveil the mechanical aspects of this behaviour such as toughness and tensile properties.

Moreover, the study highlights the impact of welding effects associated with the deposition of 20 layers, specifically in terms of heat accumulation and thermal history. The results indicate that the higher number of layers leads to increased heat retention, slower cooling rates, and a

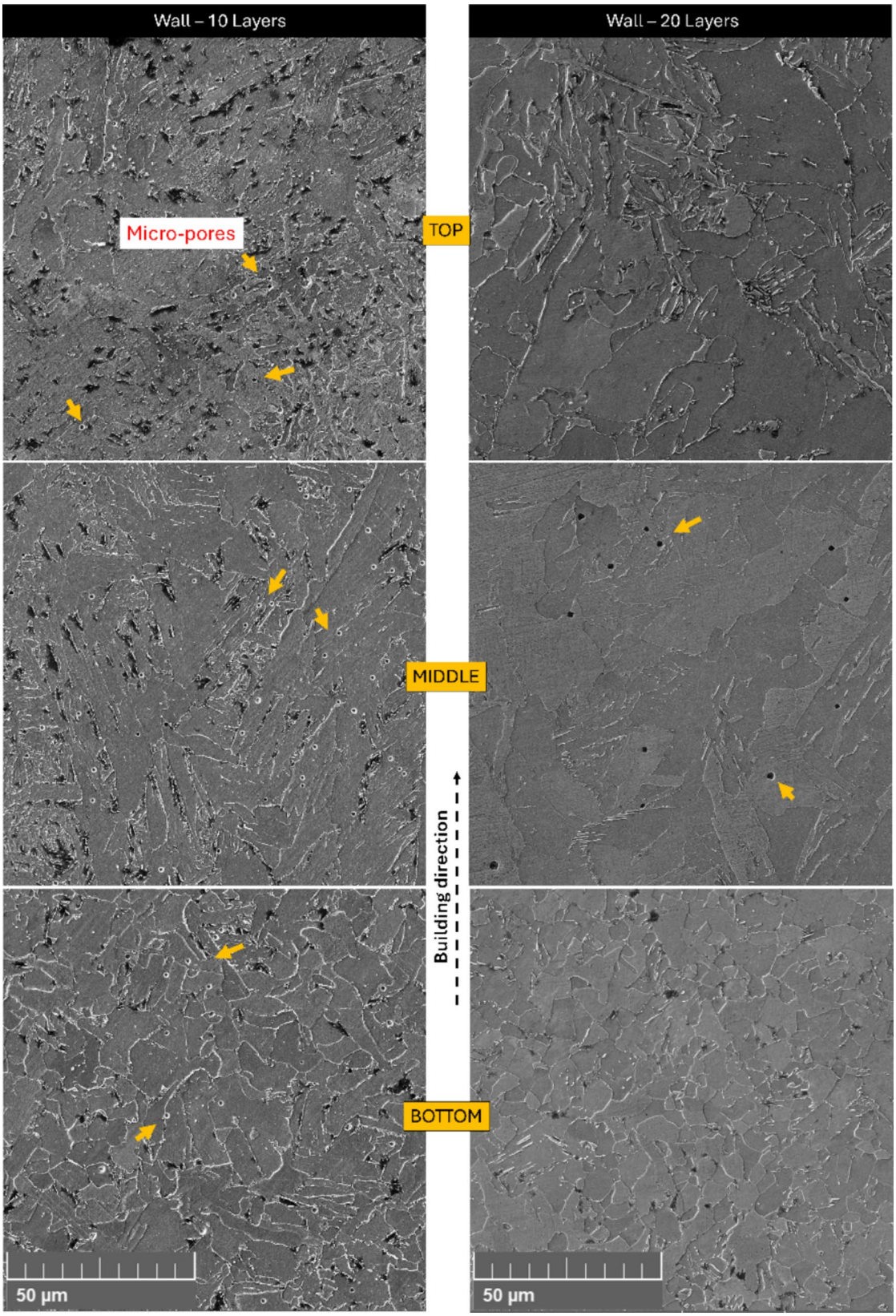


Fig. 7 The microstructure obtained from SEM for the walls deposited with 10 and 20 layers

tempering effect on previously deposited material. This is reflected in the coarser grain structures, columnar morphology, and reduced hardness in the upper sections compared to walls with fewer layers. However, there is potential to investigate these impacts further by employing alternative techniques such as diagonal printing, controlling interlayer temperature or wait time, or utilizing adaptive heat input control. These approaches could effectively mitigate heat accumulation by redistributing thermal energy and altering thermal gradients. Such modifications may promote more uniform cooling rates, leading to improved microstructural consistency and enhanced material properties.

3.4 Analysis of cubic structures

In DED-arc process, the process parameters significantly affect the deposited material. Therefore, it is crucial to understand their degree of impact on deposited structures. In this study, an optimised deposition strategy was adopted to deposit the cubic structures with various process parameter combinations to analyse their effects on hardness and microstructural evolution. The deposition strategy and the deposited cubic structures are shown in Fig. 8. For each layer, the starting position of the welding torch was rotated to 90° clockwise to achieve a homogeneous heat distribution.

From the as-deposited samples, various conclusions can be drawn since various combinations of parameters seem to result in different surface characteristics. Notably, the spatter generation, the thermal influence zone and even the size of the structures are influenced by the combined effects of the process parameters under study. For instance, the spatter islands decreased, slightly making larger cubes when the arc current was increased, or travel speed was decreased. Similarly, the thermal influence zones which are an indicator of higher heat input, decreased with increased travel speed and lower arc current.

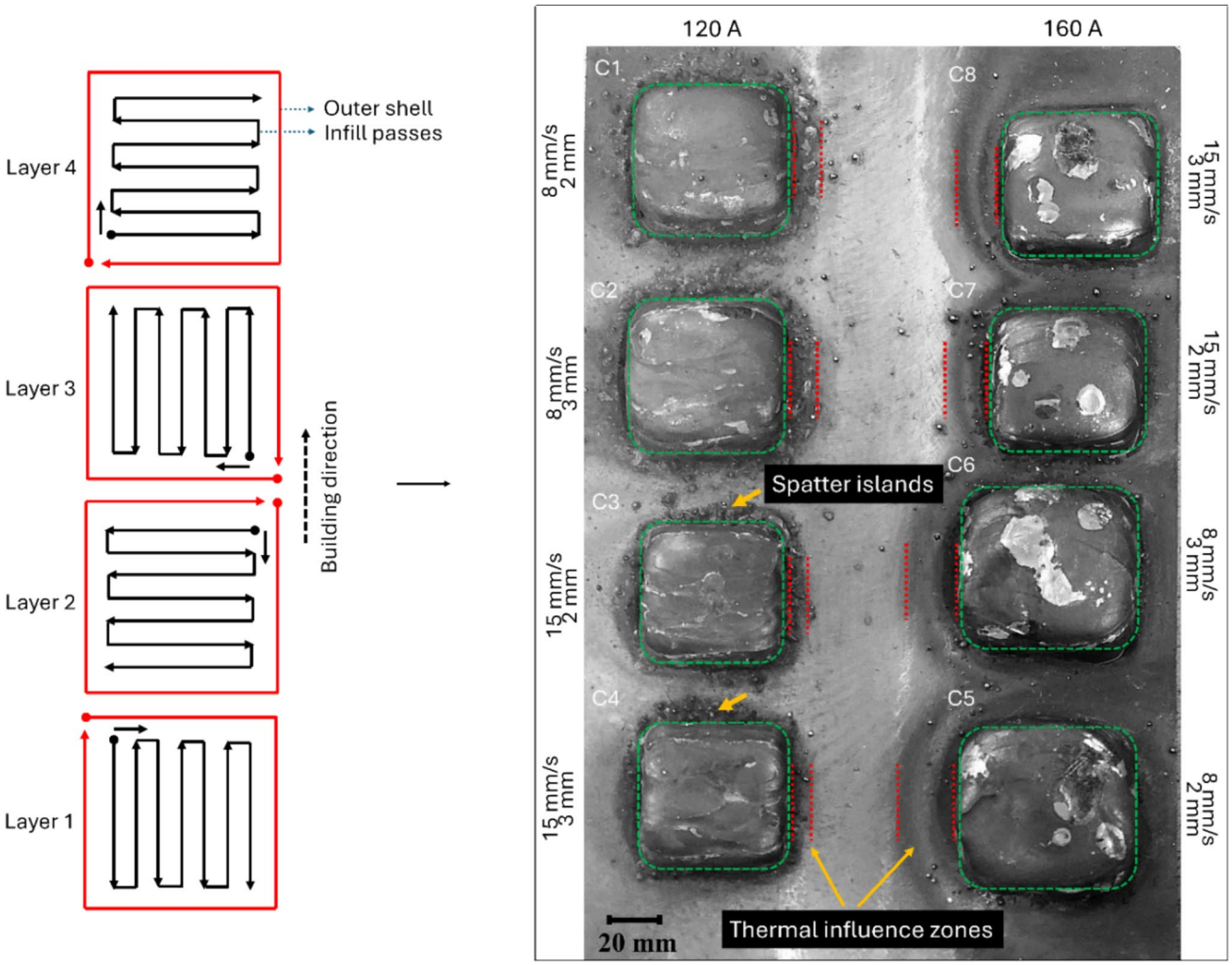


Fig. 8 The deposition strategy and the corresponding cubic structures deposited with different combinations of process parameters

The cubes with higher arc current and lower travel speed would have experienced a higher heat input, leading to larger melt-pool and exhibiting comparatively larger dimensions. That is also because in the synergic operation, when the current increases, the wire feed speed also increases, resulting in depositing more material. In addition, the cubic structures deposited with lower current and higher travel speed levels show a shallow thermal influence zone, as seen for sample C3 and C4. For a certain arc current level, increasing the travel speed decreases the heat input and the corresponding thermal influence zone. Moreover, the spatter islands are mostly seen around the samples deposited with low current level (i.e. 120 A), while the samples deposited with 160 A did not produce significant amounts of spatter. This means, the spatter generation is primarily dependent on the current level rather than the travel speed or hatch distance.

The main effects and interactions plots are shown in supplementary material ESM. 4, depicting the individual influence of each parameter and the combined effects when parameter interact with each other. The main effects plot of hardness shows that when the current is increased from 120 to 160 A, the average hardness slightly increases from 210 to 217 HV. In addition, the effects of travel speed are mor pronounced. Increasing it from 8 to 15 mm/s increased the average hardness from around 200 to 222 HV. This can be attributed to faster cooling and solidification, resulting in higher hardness. The hatch distance represents a quasi-horizontal line on the main effects plot, suggesting a very low impact on the hardness between the two levels.

The interaction plots reveal a nuanced relationship on the combined effects of parameters on hardness. Particularly, the interaction between the arc current and travel speed seems to be significant. For example, at low speed, an increase in current leads to a slight increase in hardness but at the same current level when the travel speed is increased, a substantial rise in hardness is observed. The interactions between the current and hatch distance and travel speed and hatch distance are not that significant. Thus, under these parameter levels, the higher current and higher hardness results in higher hardness in the cubic structures.

The hardness profiles from top to bottom of each sample and their average values are shown in Fig. 9. On the left, the hardness profiles illustrate the variability of hardness for each cubic sample (C1 to C8) in the building direction. Although the data shows small fluctuations in hardness across the wall sections, the overall trend remains fairly consistent from top to bottom, indicating a well-controlled deposition. The small variations, although not so significant, have might been induced as a result of local variations in the cooling rates during the solidification or because of small microstructural inhomogeneity. These hardness profiles suggest that the adopted deposition strategy proved to be effective in producing uniform hardness across all the samples.

The bar charts on the right side of the figure gives a detailed analysis by comparing the average hardness for each sample. The hardness values are relatively close to each other, ranging from 191 to 225 HV to be the minimum and maximum average values among the samples. Although small variations of average hardness exist at different current levels; however, the driving force seems to be the travel speed for these samples while the underlying hatch distances seem to have negligible influence on the hardness. The highest average hardnesses were observed for samples C3, C4, C7 and C8. C3 and C4 were deposited with 120 A current while C7 and C8 with 160 A. The main point for these samples lies in the travel speeds; all of them were deposited with

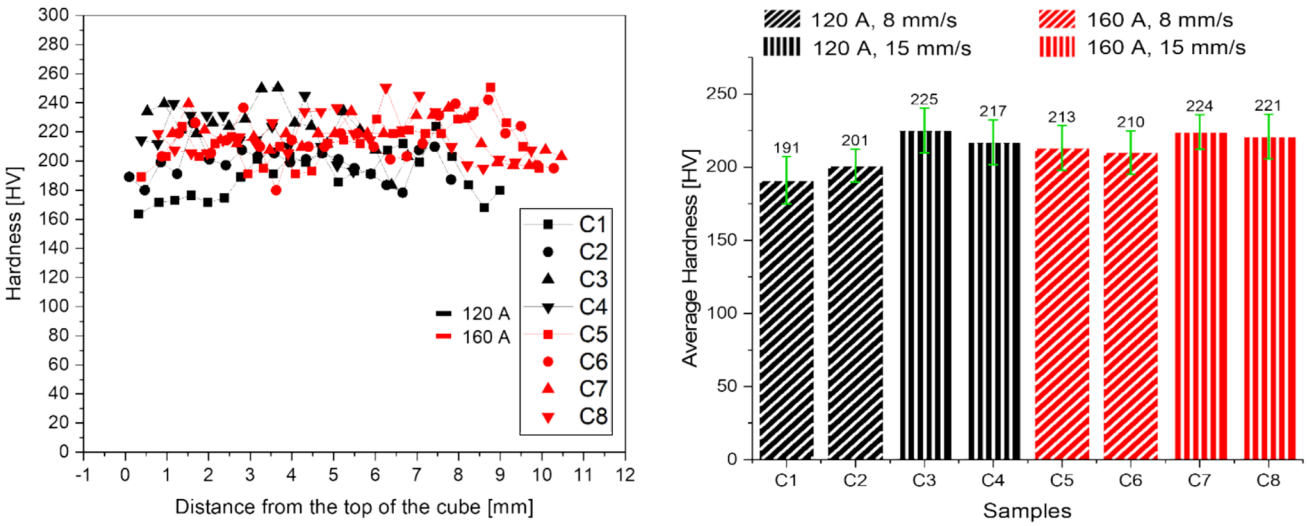


Fig. 9 Hardness profiles and average hardness of all cubic structures deposited with various parameters

15 mm/s. Thus, higher travel speed favours in producing a harder material. Such results were observed in another study where higher travel speeds resulted in higher average hardness [41]. Moreover, the lowest hardness was observed for sample C1 that corresponds to 120 A and 8 mm/s of travel speed.

The optical microscopic images of sample C1 to C8 are shown in Fig. 10, demonstrating the microstructure at the centre of each sample. The microstructure primarily exhibits a ferritic-pearlitic structure that is typically associated to low-carbon steels such as ER70S-6. All the microstructures denote a mixture of pearlite and ferrite, which are represented by dark and lighter regions, respectively. The microstructure primarily presents a clean and homogeneous grain structure with lamellar appearance, representing needle-shaped lamellae. However, the concentration and length of cementite lamellae may offer slight differences across various samples. For instance, samples C3, C4, C7 and C8 apparently present a higher lamellae density, leading to a slightly finer microstructure. This could be associated to the higher material hardness for these samples. However, these differences are not very significant, and the microstructure across all samples seem to be very similar. Therefore, the underlying deposition strategy used for the cubic structures is effective in achieving a homogeneous microstructure with a uniform material hardness.

The optical microscopic images showed comparable results in the centre of each sample. However, to fully understand the microstructural features in the building direction, SEM images were obtained at three locations—top, middle and bottom sections of sample C4, shown in ESM. 4 as supplementary material. The top section shows a dominance of a columnar grain structure in which the grains seem to have grown preferentially towards the direction of heat dissipation i.e. towards the substrate. The presence of these elongated grain structure suggests that the top section has undergone a comparatively slower cooling, promoting the evolution of columnar structures. The grain boundaries are well-defined and show a homogeneous distribution which can be beneficial for mechanical properties. The middle section of the sample shows a similar lamellar structure but with shorter lamellae compared to the top section. There is also the evidence of some micro-pores. However, their size is small and have a low concentration. Finally, the bottom consists of a finer polygonal and equiaxed structure which is explicitly different from the columnar structure observed in the top and middle sections. The formation of polygonal structure can be associated with the structural reheating as a result of subsequent layering [42]. The bottom region directly facing the colder substrate further promotes such a microstructure by a corresponding faster cooling. However, as observed in the case of single-pass multilayer walls, when the number of layers increase, the intense thermal cycles tend to reshape the microstructure by inducing a coarser polygonal structure.

Concluding the overall SEM results, the microstructure across different sections consists of a mixture of ferrite and pearlite phases. The ferrite matrix with randomly oriented pearlite colonies indicates a moderate cooling during the deposition. Moreover, the variations in grain size and orientation are indicative of small differences in the thermal history in the building direction. However, the overall microstructure presents a homogeneous and uniform microstructure using this deposition strategy.

3.5 Conclusions

The study concludes the following main findings.

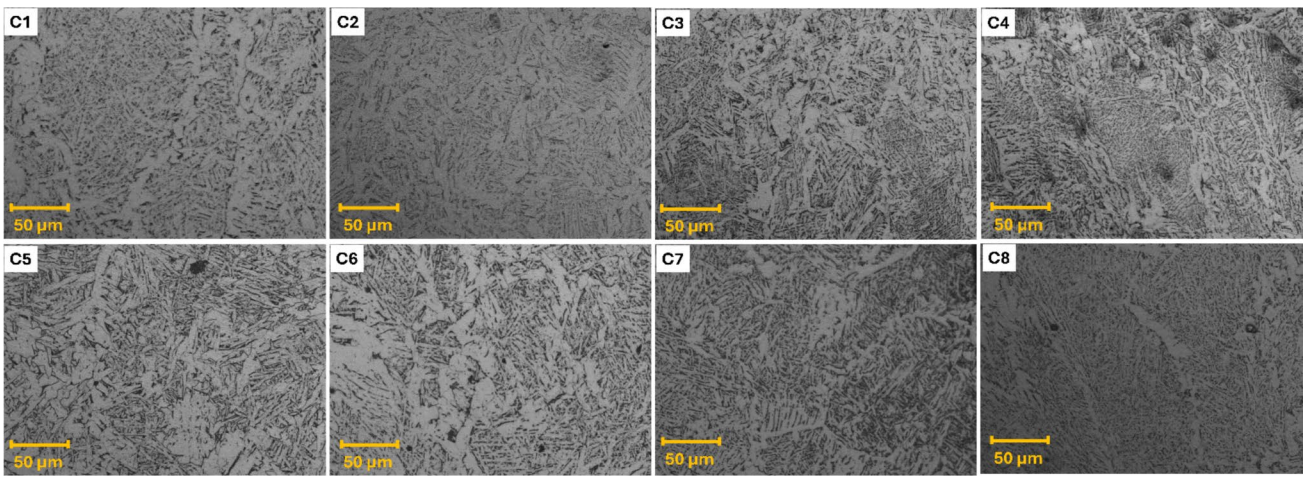


Fig. 10 Microstructure of the cubic structures captured in the centre of each sample

- The bead-on-plate analysis revealed a positive correlation of arc current with bead characteristics and an inverse relationship with travel speed.
- The interactions of factors showed that the influence of current becomes less pronounced at high travel speeds and travel speed seems to be the driving force even at higher current levels.
- For single-pass multilayer walls the unidirectional strategy resulted in a wall with a steep slop; producing an uneven wall with varied wall height. This was effectively resolved by either adopting a bidirectional path strategy or the unidirectional deposition strategy integrated with a dwell time.
- The walls with 10 and 20 layers both exhibit a clear transition from a polygonal to a columnar structure after the 5th layer. Both walls represented a columnar structure with some acicular and Widmanstätten ferrites at the top, partly polygonal grains in the middle while a polygonal structure at the bottom.
- Compared to 10-layer wall, the wall deposited with 20 layers showed a coarser microstructure in the middle and bottom sections which is also reflected to their hardness profiles. The 10-layer wall exhibited a slightly higher average hardness.
- An increase of arc current and travel speed slightly increased the hardness in cubic structures; however, this is more pronounced at higher travel speed that is reflected in higher lamellae density in the microstructure.
- The SEM revealed a transition from a lamellar to a polygonal structure from top to bottom of the cubic sample.

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Author contributions Hambal Iqbal: involved in experiments, investigations, data analysis, and writing – original draft. Alessandro Ascari: conceptualization, validation and supervising the experiments. Erica Liverani: validated microstructural studies, SEM analysis and supervised the work. Alessandro Fortunato: coordinated research activities, provided experimental facilities and supervision. All the authors conducted reviews and editing of initial draft.

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Data availability The data collected and analysed in this study are presented as figures within this study. The additional data and information from this work can be shared upon request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Ethical approval The authors state that the research did not involve vulnerable participants, minors, clinical patients, human tissue, animals, human biological material, and genetically modified organisms.

Consent to participate The authors understand that the manuscript and associated personal data will be shared with Research Square for the delivery of the author dashboard.

Consent for publication The authors agree with the Copyright Transfer Statement.

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